

UteroCare

Medical Knowledge Base — Part 1

Clinical Reference for the AI Clinical Assistant

Prepared for healthcare professionals, medical students, and the UteroCare RAG knowledge system

Reference Edition — 2026

1. About UteroCare

1.1 Project Overview

UteroCare is a clinical decision-support platform built to assist in the early identification and risk stratification of uterine pathology, with an initial focus on uterine fibroids (leiomyomas). The system combines pelvic MRI image analysis with structured patient-reported symptom data to generate an integrated prediction that supports — but does not replace — clinical judgment.

The platform is intended for use alongside standard gynecological evaluation, providing clinicians with a consistent, reproducible second layer of analysis and giving patients an accessible explanation of how a given prediction was reached.

1.2 Objective of the System

- Support earlier flagging of patients who may benefit from further gynecological work-up.
- Reduce variability in the interpretation of overlapping symptom presentations.
- Provide a transparent, explainable output rather than an opaque risk number.
- Offer a conversational AI Clinical Assistant that can answer follow-up questions about a prediction in plain clinical language.

1.3 AI Workflow

Predictions are produced through a five-stage pipeline. Each stage passes structured output to the next, and the final stage is responsible for communicating the result.

Stage	Function
1. MRI Model	Analyzes T1/T2-weighted pelvic MRI sequences to detect, localize, and characterize uterine lesions (size, margins, signal intensity, enhancement pattern).
2. Symptom Model	Processes structured patient-reported symptoms (bleeding pattern, pain, pressure symptoms, history) to estimate a symptom-based likelihood profile.
3. Fusion Prediction	Combines imaging-derived and symptom-derived outputs into a single multimodal prediction using a weighted fusion layer.
4. Risk Assessment	Converts the fused prediction into a categorical risk tier (e.g., low / moderate / high) using calibrated thresholds and relevant clinical modifiers (age, anemia status, growth rate).
5. AI Assistant	Translates the numerical/categorical output into a natural-language explanation for the clinician or patient, drawing on this knowledge base.

1.4 Purpose of the AI Clinical Assistant

The AI Clinical Assistant is the natural-language interface of UteroCare. Its role is to explain — clearly and accurately — why the system produced a given prediction, which inputs contributed most, and what a reasonable next clinical step might be. It is a communication layer, not an independent diagnostic authority: its explanations are always secondary to, and should always defer to, the assessment of a licensed clinician.

Section Summary

- UteroCare integrates MRI and symptom data through a five-stage AI pipeline ending in an explainable output.
- The system is a decision-support tool, not a replacement for clinical evaluation.
- The AI Clinical Assistant's core function is transparent explanation, not autonomous diagnosis.

2. Female Reproductive System

2.1 Basic Anatomy of the Uterus

The uterus is a pear-shaped, muscular pelvic organ situated between the bladder anteriorly and the rectum posteriorly. It is anatomically divided into the fundus (the domed upper portion above the tubal insertion points), the corpus (the main body), the isthmus (a narrowed transitional segment), and the cervix (the lower fibromuscular portion opening into the vagina). The uterine wall is composed of three concentric layers: the perimetrium (outer serosal covering), the myometrium (middle smooth-muscle layer), and the endometrium (inner mucosal lining).

2.2 Endometrium

The endometrium is the mucosal lining of the uterine cavity and consists of a functional layer, which thickens and is shed cyclically under hormonal influence, and a basal layer, which regenerates the functional layer after each cycle. Endometrial thickness and echotexture are routinely assessed on ultrasound and MRI and are important in distinguishing endometrial pathology (e.g., polyps, hyperplasia) from myometrial pathology such as fibroids.

2.3 Myometrium

The myometrium is the thick middle layer of smooth muscle responsible for uterine contractions during labor and menstruation. It is subdivided into inner, middle, and outer muscular zones. Uterine fibroids arise from the smooth muscle cells of the myometrium and are classified in part by their position relative to these layers and the endometrial cavity.

2.4 Cervix

The cervix is the cylindrical lower segment of the uterus, extending from the internal os (opening into the uterine cavity) to the external os (opening into the vagina). It is composed primarily of fibrous connective tissue with a smaller smooth-muscle component and is lined by columnar epithelium in the endocervical canal and squamous epithelium on the ectocervix.

2.5 Ovaries

The ovaries are paired almond-shaped gonads located in the ovarian fossae on the lateral pelvic walls. They house the oocyte reserve, support follicular development, and secrete estrogen and progesterone, the principal hormones driving the menstrual cycle and endometrial changes.

2.6 Fallopian Tubes

The fallopian tubes extend from the superolateral uterine cornua toward the ovaries and are divided into the interstitial, isthmic, ampullary, and fimbrial segments. The ampulla is the typical site of fertilization, and ciliary action along the tubal epithelium transports the oocyte or early embryo toward the uterine cavity.

2.7 Menstrual Cycle

The menstrual cycle averages 28 days (with a normal range of roughly 21–35 days) and is divided into the menstrual, proliferative, and secretory phases, with ovulation marking the transition between the proliferative and secretory phases.

Phase	Approx. Days	Dominant Hormone	Endometrial Change
Menstrual	Days 1–5	Low estrogen/progesterone	Shedding of the functional layer
Proliferative	Days 6–13	Rising estrogen	Regrowth and thickening of the endometrium
Ovulation	Day ~14	LH surge	Endometrium continues to thicken
Secretory	Days 15–28	Progesterone (with estrogen)	Glandular secretion; endometrium prepares for implantation

2.8 Hormonal Regulation

The menstrual cycle is governed by the hypothalamic–pituitary–ovarian (HPO) axis. The hypothalamus releases gonadotropin-releasing hormone (GnRH) in a pulsatile manner, stimulating the anterior pituitary to secrete follicle-stimulating hormone (FSH) and luteinizing hormone (LH). FSH promotes follicular development and estrogen production; the mid-cycle LH surge triggers ovulation. After ovulation, the corpus luteum secretes progesterone, which stabilizes and matures the endometrium; if implantation does not occur, corpus luteum regression leads to a fall in progesterone and menstruation.

Section Summary

- The uterus has three tissue layers — perimetrium, myometrium, endometrium — each with distinct clinical relevance.
- Fibroids originate in the myometrium; endometrial changes are cyclical and hormone-dependent.
- The HPO axis (GnRH → FSH/LH → estrogen/progesterone) governs the menstrual cycle and is a common target of hormonal fibroid therapy.

3. Common Uterine Symptoms

The following symptoms are frequently reported in patients undergoing evaluation for uterine pathology, including fibroids. Each entry summarizes its definition, common causes, and clinical significance for triage and diagnostic reasoning.

Heavy menstrual bleeding

Definition: Menstrual blood loss that is subjectively excessive or objectively exceeds approximately 80 mL per cycle, often with frequent pad/tampon changes or passage of clots.

Possible causes: Fibroids (especially submucosal/intramural), adenomyosis, endometrial polyps, coagulation disorders, hormonal imbalance, endometrial hyperplasia.

Clinical significance: A leading indication for fibroid work-up; strongly associated with iron-deficiency anemia and reduced quality of life.

Pelvic pain

Definition: Discomfort or pain localized to the lower abdomen or pelvis, which may be cyclical or constant.

Possible causes: Fibroid degeneration, adenomyosis, endometriosis, pelvic inflammatory disease, ovarian cysts.

Clinical significance: Character and timing (cyclic vs. constant) help differentiate fibroid-related pain from other gynecological or non-gynecological causes.

Menstrual cramps (dysmenorrhea)

Definition: Cramping lower abdominal pain occurring just before or during menstruation.

Possible causes: Prostaglandin-mediated uterine contractions, fibroids, adenomyosis, endometriosis.

Clinical significance: Severity out of proportion to typical primary dysmenorrhea warrants imaging to exclude structural pathology.

Bleeding between periods (intermenstrual bleeding)

Definition: Uterine bleeding occurring outside the expected menstrual window.

Possible causes: Endometrial polyps, submucosal fibroids, hormonal contraception, cervical pathology, endometrial hyperplasia.

Clinical significance: Requires evaluation to exclude structural or premalignant endometrial causes, particularly if persistent.

Bleeding after menopause (postmenopausal bleeding)

Definition: Any vaginal bleeding occurring 12 months or more after the final menstrual period.

Possible causes: Endometrial atrophy (most common), endometrial polyps, fibroids, endometrial hyperplasia or carcinoma.

Clinical significance: Considered a red-flag symptom; endometrial carcinoma must be excluded in every case.

Periods longer than 7 days (prolonged menstruation)

Definition: Menstrual bleeding lasting longer than the typical 4–7 day duration.

Possible causes: Submucosal fibroids, adenomyosis, hormonal dysregulation, anovulatory cycles.

Clinical significance: Often coexists with heavy bleeding and increases risk of anemia; suggests structural rather than purely functional cause.

Pain during intercourse (dyspareunia)

Definition: Pain occurring during or after sexual intercourse.

Possible causes: Large or posteriorly located fibroids, adenomyosis, endometriosis, pelvic adhesions.

Clinical significance: Deep dyspareunia is more suggestive of structural pelvic pathology than superficial causes.

Frequent urination

Definition: Increased urinary frequency, often with a sense of incomplete bladder emptying.

Possible causes: Anterior uterine fibroids compressing the bladder, large fibroid uterus, pregnancy.

Clinical significance: A common bulk-related symptom of anterior or large fibroids; helps localize mass effect on imaging correlation.

Constipation

Definition: Infrequent or difficult defecation.

Possible causes: Posterior uterine fibroids compressing the rectosigmoid colon, large pelvic mass effect.

Clinical significance: A bulk-related symptom that supports posterior fibroid localization when correlated with imaging.

Pelvic pressure

Definition: A sensation of fullness, heaviness, or bulk in the lower pelvis.

Possible causes: Enlarging fibroids, large fibroid uterus, adenomyosis, pelvic mass.

Clinical significance: Correlates with uterine or fibroid volume and is commonly used to gauge symptom burden over time.

Abdominal swelling / distension

Definition: Visible or palpable enlargement of the lower abdomen.

Possible causes: Large or multiple fibroids, significantly enlarged uterus, less commonly ovarian mass.

Clinical significance: Rapid or marked distension warrants imaging to assess growth rate and exclude alternative pelvic or abdominal pathology.

Lower back pain

Definition: Pain localized to the lumbosacral region.

Possible causes: Posterior fibroids, large uterine masses exerting pressure on pelvic nerves or structures, adenomyosis.

Clinical significance: Non-specific in isolation but relevant when it correlates temporally with other bulk-related symptoms.

Abnormal vaginal discharge

Definition: Discharge that differs from a patient's normal pattern in volume, color, odor, or consistency.

Possible causes: Infection, cervical pathology, submucosal fibroid necrosis or ulceration, endometrial pathology.

Clinical significance: Foul-smelling or blood-tinged discharge in the context of a known fibroid may suggest degeneration or infection of a submucosal lesion.

Fatigue

Definition: Persistent tiredness or reduced energy not relieved by rest.

Possible causes: Chronic blood loss leading to iron-deficiency anemia, underlying systemic conditions.

Clinical significance: Frequently secondary to heavy menstrual bleeding; should prompt a hemoglobin/ferritin check.

Diagnosed anemia

Definition: Laboratory-confirmed reduction in hemoglobin or hematocrit below the normal reference range.

Possible causes: Chronic heavy or prolonged menstrual bleeding, nutritional deficiency.

Clinical significance: A key objective marker of symptom severity in fibroid-related bleeding; influences urgency of treatment.

Difficulty conceiving (subfertility)

Definition: Inability to conceive after 12 months of regular unprotected intercourse (or 6 months if the patient is over 35).

Possible causes: Submucosal or cavity-distorting fibroids, tubal or endometrial factors, adenomyosis, other causes of infertility.

Clinical significance: Fibroid location relative to the endometrial cavity — rather than size alone — is the primary determinant of fertility impact.

Section Summary

- Symptoms cluster into bleeding-pattern symptoms, bulk/pressure symptoms, and pain symptoms — each pointing toward different fibroid locations or alternative diagnoses.
- Postmenopausal bleeding is always a red-flag symptom requiring exclusion of malignancy.
- Fibroid location relative to the endometrial cavity matters more than size for fertility-related counseling.

4. Uterine Fibroids

4.1 Overview

Uterine fibroids (leiomyomas) are benign, monoclonal tumors arising from the smooth muscle cells of the myometrium. They are the most common pelvic tumor in women of reproductive age, with prevalence increasing with age until menopause, after which most fibroids regress due to declining estrogen levels.

4.2 Causes

- Somatic genetic alterations in myometrial cells, most notably MED12 mutations, driving clonal smooth-muscle proliferation.
- Sensitivity to estrogen and progesterone, which promote fibroid growth.
- Dysregulated extracellular matrix production leading to the dense fibrous component characteristic of these tumors.

4.3 Risk Factors

Risk Factor	Association
Age approaching menopause	Cumulative hormonal exposure increases prevalence until the perimenopausal years.
Family history	First-degree relatives with fibroids confer increased risk.
African ancestry	Associated with earlier onset, larger size, and higher symptom burden.
Nulliparity	Lower parity is associated with increased risk.
Obesity	Higher body mass index correlates with increased circulating estrogen and fibroid risk.
Early menarche	Longer lifetime estrogen exposure increases risk.
Vitamin D deficiency / hypertension	Both are reported as modifiable or associated risk factors in current literature.

4.4 Symptoms

Symptom presentation depends heavily on fibroid location, size, and number. Common presentations include heavy or prolonged menstrual bleeding, pelvic pressure or pain, urinary frequency, constipation, dyspareunia, and, in cavity-distorting lesions, subfertility or recurrent pregnancy loss. Many fibroids remain entirely asymptomatic and are identified incidentally.

4.5 MRI Findings

MRI is the most accurate imaging modality for fibroid mapping, particularly for pre-treatment planning. Typical findings include:

- Well-circumscribed masses that are hypointense (dark) relative to myometrium on T2-weighted sequences.
- Homogeneous signal in typical fibroids; heterogeneous or hyperintense areas suggest degeneration (hyaline, cystic, or red/hemorrhagic degeneration).

- Enhancement pattern on contrast-enhanced sequences that generally parallels the surrounding myometrium, which helps distinguish typical fibroids from leiomyosarcoma.
- Location classified using the FIGO submucosal-to-subserosal system (see comparison table below).
- Restricted diffusion or atypical enhancement, combined with rapid interval growth, should raise suspicion for a rare malignant leiomyosarcoma and prompt further work-up.

FIGO Classification of Fibroid Location

Type	Location	Clinical Note
0	Pedunculated intracavitary (entirely within the endometrial cavity)	Most amenable to hysteroscopic resection; strong impact on bleeding and fertility.
1	Submucosal, <50% intramural extension	Frequently causes heavy bleeding; generally hysteroscopically resectable.
2	Submucosal, ≥50% intramural extension	May require combined hysteroscopic/laparoscopic approach.
3	Intramural, contacts endometrium but does not distort cavity on imaging	Can still affect implantation despite not visibly distorting the cavity.
4	Intramural, entirely within the myometrium	Common; symptom impact depends on size and number.
5–7	Subserosal, with varying intramural component down to pedunculated	Typically bulk/pressure symptoms rather than bleeding.
8	Other (e.g., cervical, broad ligament, parasitic)	Location-specific symptoms; may complicate surgical planning.

Comparison: Fibroids vs. Adenomyosis vs. Endometrial Polyps

Feature	Fibroid	Adenomyosis	Endometrial Polyp
Origin	Myometrial smooth muscle	Ectopic endometrial glands within myometrium	Endometrial tissue overgrowth
Margins on imaging	Well-circumscribed	Poorly defined, diffuse junctional zone thickening	Well-defined, intracavitary
Typical T2 signal	Hypointense	Heterogeneous with ill-defined myometrial thickening	Intermediate, similar to endometrium
Bleeding pattern	Heavy, sometimes prolonged	Heavy, painful menstruation	Intermenstrual spotting common
Uterine contour	Often distorted/enlarged, asymmetric	Globular, symmetric enlargement	Usually normal contour

4.6 Diagnosis

- Bimanual pelvic examination to assess uterine size, contour, and mobility.
- Transvaginal ultrasound as the first-line imaging modality for initial detection.
- Pelvic MRI for precise mapping, surgical planning, or when ultrasound findings are inconclusive.

- Hysteroscopy for direct visualization and management of submucosal lesions.
- Complete blood count to assess for anemia secondary to chronic blood loss.
- Exclusion of malignancy in atypical presentations (rapid growth, postmenopausal new fibroid, atypical MRI features).

4.7 Treatment

Category	Options
Expectant management	Observation with periodic imaging for asymptomatic or minimally symptomatic fibroids.
Medical therapy	GnRH agonists/antagonists, progestins, tranexamic acid, NSAIDs, combined hormonal contraceptives for symptom control.
Minimally invasive procedures	Uterine artery embolization, MRI-guided focused ultrasound surgery, hysteroscopic or laparoscopic myomectomy.
Surgical management	Open (abdominal) myomectomy for large or multiple fibroids; hysterectomy for definitive treatment when fertility preservation is not required.

4.8 Lifestyle Advice

- Maintain a healthy body weight, as adiposity is associated with higher circulating estrogen levels.
- Ensure adequate iron and vitamin D intake, particularly in patients with heavy menstrual bleeding.
- Engage in regular physical activity as tolerated.
- Keep a symptom diary (bleeding pattern, pain, bulk symptoms) to support monitoring between visits.

4.9 Follow-Up Care

- Periodic ultrasound or MRI surveillance to monitor fibroid size and growth rate.
- Regular hemoglobin/ferritin checks in patients with a history of heavy bleeding.
- Reassessment of fertility goals and cavity involvement in patients desiring pregnancy.
- Prompt re-evaluation for rapid growth, new postmenopausal symptoms, or treatment-resistant bleeding.

Section Summary

- Fibroids are benign but the leading cause of structural abnormal uterine bleeding in reproductive-age women.
- FIGO location classification, not size alone, drives symptom pattern and treatment choice.
- Management ranges from expectant observation to definitive surgery, tailored to symptom severity and fertility goals.

5. AI Prediction Explanation Templates

The templates below are reference structures the AI Clinical Assistant uses to generate consistent, professional explanations of model outputs. Bracketed placeholders (e.g., {risk_score}) represent values to be populated dynamically from the pipeline output described in Section 1.3.

5.1 "Why was Fibroid predicted?"

"Based on the analysis of your pelvic MRI, the model identified {finding_summary} (e.g., a well-defined, T2-hypointense mass measuring {size} cm in the {location} region). This imaging pattern is consistent with a uterine fibroid. In combination with your reported symptoms ({symptom_list}), the fusion model assigned a fibroid likelihood of {probability}%."

This explanation should always name the imaging evidence first, then connect it to symptom evidence, mirroring standard clinical reasoning.

5.2 "Why is my risk score high?"

"Your risk score of {risk_score} ({risk_tier}) reflects a combination of factors: {top_contributing_factors}. Factors such as {clinical_modifiers} (e.g., prolonged bleeding duration, confirmed anemia, or rapid interval growth) increased the weighting of this assessment. This score reflects statistical risk stratification and should be interpreted alongside a full clinical evaluation."

The explanation must avoid presenting the score as a diagnosis and should always frame it as a decision-support input.

5.3 "Which symptoms influenced the prediction?"

"The symptom model gave the greatest weight to: {ranked_symptom_list}. These symptoms are commonly associated with the predicted condition because {brief_clinical_rationale}. Symptoms you reported that were not strongly weighted include {low_weight_symptoms}, as they are less specific to this condition."

Ranking transparency helps clinicians validate that the model's reasoning aligns with clinical expectations.

5.4 "What MRI findings contributed to the result?"

"The imaging model identified the following features: lesion location ({figo_type}), size ({size} cm), signal characteristics ({signal_description}), and enhancement pattern ({enhancement_pattern}). These findings are most consistent with {predicted_finding} rather than alternative diagnoses such as {differential_list}."

This template should reference the FIGO classification and comparison criteria in Section 4.5 for consistency.

5.5 "What does the confidence score mean?"

"The confidence score of {confidence_score}% reflects the model's internal certainty in this specific prediction, based on the clarity and consistency of the input data. A higher confidence score indicates that the imaging and symptom data strongly align with the predicted pattern; a lower score indicates greater

uncertainty and suggests that additional clinical correlation or repeat imaging may be valuable."

Confidence must never be equated with clinical severity — a low-severity finding can have high confidence, and vice versa.

5.6 "What should the patient do next?"

"Based on this result, a reasonable next step is {recommended_action} (e.g., referral to a gynecologist for confirmatory evaluation, a complete blood count to assess for anemia, or discussion of treatment options outlined in Section 4.7). This recommendation is intended to support – not replace – a full consultation with your treating clinician."

Every "next steps" response must end with an explicit statement that the assistant's output supports, and does not substitute for, clinician judgment.

Section Summary

- Every explanation template links model output back to specific, named evidence (imaging findings, symptoms, or modifiers).
- Risk and confidence scores must always be framed as decision-support metrics, never as diagnoses.
- All patient-facing "next steps" language must defer explicitly to clinician judgment.